BETAMETHASONE-G TABLET 0.5MG

Ingredient(s):
Each tablet contains:
Betamethasone 0.5mg

Pharmacology (Summary of Pharmacodynamic and Pharmacokinetics):

1. Betamethasone is a glucocorticosteroid. Betamethasone is a synthetic derivative of prednisolone which is about eight to ten times as active as prednisolone on a weight-for-weight basis. Betamethasone is an anti-inflammatory adrenal corticosteroid. It provides potent anti-inflammatory, antirheumatic and antiallergic effects in the treatment of corticosteroid-responsive disorders. Glucocorticosteroids, eg. betamethasone, cause profound and varied metabolic effects and modify the body's immune response to diverse stimuli. Betamethasone has high glucocorticosteroid activity and slight mineralocorticosteroid activity.
2. Corticosteroids are bound to plasma proteins in varying degrees. Corticosteroids are metabolised primarily in the liver and are then excreted by the kidneys.

Indication(s):
Management of various endocrine, musculoskeletal, collagen, dermatologic, allergic, ophthalmic, gastrointestinal, respiratory, hematologic, neoplastic and other diseases known to be responsive to corticosteroid therapy. Corticosteroid hormone therapy is an adjunct to, and not a replacement for, conventional therapy.

Dosage and Administration:
Dosing requirements are variable and must be individualized on the basis of the specific disease, its severity and the response of the patient.
The initial dose of betamethasone tablets may vary from 0.25mg to 8mg daily depending on the specific disease being treated. Generally low doses will suffice in situations of less severity. However, selected patients may require higher initial doses. This initial dose should be maintained or adjusted until a satisfactory response is observed.
If a satisfactory clinical response does not occur after a reasonable period of time, betamethasone should be discontinued and the patient transferred to other appropriate therapy.
The usual initial pediatric oral dosage varies from 17.5-250mcg/kg body weight/day or 0.5-7.5mg/m^2 of body surface/day. Dosages for infants and children should be governed by the same considerations as adults rather than strict adherence to ratios indicated by age or body weight.
When a favourable response is reached, the proper maintenance dose should be determined by decreasing the initial dose in small decrements at appropriate time intervals until the lowest dose, which will maintain an adequate clinical response, is reached.
Treatment should be discontinued if a period of spontaneous remission occurs in a chronic condition.
Exposure of the patient to stressful situations unrelated to the disease under treatment may necessitate an increase in the dosage of betamethasone. If betamethasone is to be discontinued after long term therapy, dosage should be decreased gradually.
Once-A-Day Dosage: As a convenience to the patient and to insure improved compliance with dosing, the total daily maintenance dose can be administered once in the morning.
Alternate Day Therapy: Betamethasone is <u>not</u> recommended for alternate-day dosing due to its long biological half-life (36 to 54 hours) with associated suppressive effects on the HPA axis. If oral long term use is required for disease therapy, an alternate day dosing regime with an intermediate acting adrenocorticosteroid (such as prednisone, prednisolone or methylprednisolone) should be considered.
To be dispensed on physician's prescription.

Contraindication(s):
Betamethasone is contraindicated in patients with systemic fungal infections and those who have shown hypersensitivity to betamethasone or other corticosteroids, or to any of the components of betamethasone tablets.

Side Effect(s) / Adverse Reaction(s):
Adverse reactions to betamethasone are the same as those reported for other corticosteroids and relate both to dose and duration of therapy. These reactions can usually be reversed or minimized by a dosage reduction. Generally, this is preferable to drug treatment withdrawal.
Side effects of betamethasone include fluid and electrolyte disturbances, muscle weakness, osteoporosis, peptic ulcer with possible perforation and hemorrhage, impaired wound healing, facial erythema, convulsions, menstrual irregularities suppression of growth in children, decreased carbohydrate tolerance, increased intraocular pressure and negative nitrogen balance due to protein catabolism. Prolonged treatment with corticosteroids in high dosage is occasionally associated with subcapsular cataract and skin thinning.

Precaution(s) / Warning(s):

1. Remission or exacerbation of the disease process, the patient's individual response to therapy and exposure to emotional or physical stress, such as serious infection, surgery or injury may necessitate dosage adjustments. Monitoring may be required for up to 1 year following cessation of long-term or high-dose corticosteroid therapy.
2. Some signs of infection may be masked by corticosteroids and new infections may appear during use. There may be decreased resistance and inability to localize infection when corticosteroids are used.
3. Prolonged corticosteroid use may produce posterior subcapsular cataracts (especially in children), glaucoma with possible damage to the optic nerves and may enhance secondary ocular infections due to fungi or viruses.
4. Average and large doses of corticosteroids can cause elevation of blood pressure, salt and water retention and increased excretion of potassium. However, these effects are less likely to occur with the synthetic derivatives except when used in large doses. Dietary salt restriction and potassium supplementation may be considered. All corticosteroids increase calcium excretion.
5. Patients should not be vaccinated against smallpox while on corticosteroid therapy. Other immunization procedures should not be undertaken in patients receiving corticosteroids, especially high doses, because of possible hazards of neurological complications and lack of antibody response. However, immunization procedures may be undertaken in patients that are receiving corticosteroids as replacement therapy eg. for Addison's disease.
6. Patients who are on immunosuppressant doses of corticosteroids should be warned to avoid exposure to chickenpox or measles. If exposed, to obtain medical advice.
7. Corticosteroid therapy in active tuberculosis should be restricted to those cases of fulminating or disseminated tuberculosis in which the corticosteroid is used for management in conjunction with appropriate antituberculosis regime.
8. If corticosteroids are indicated in patients with latent tuberculosis or tuberculin activity, close observation is necessary since reactivation of the disease may occur. Chemoprophylaxis should be given to patients with prolonged corticosteroid therapy. If rifampin is used in chemoprophylactic program, its enhancing effect on metabolic hepatic clearance of corticosteroids should be considered; adjustment in corticosteroid dosage may be required.

9. The lowest possible dose of corticosteroid should be used to control the condition under treatment. Whenever possible, dosage should be reduced gradually.
10. Too rapid corticosteroid withdrawal may result in drug-induced secondery adrenocortical insufficiency. This may be minimized by gradual dosage reduction. Such relative insufficiency may persist for months after discontinuation of therapy. Therefore corticotherapy should be reinstituted if stress occurs during that period. Dosage may have to be increased if the patient is receiving corticosteroids. Since mineralocorticoid secretion may be impaired, salt and/or a mineralocorticosteroid should be given concurrently.
11. Corticosteroid effect is enhanced in patients with hypothyroidism or in those with cirrhosis.
12. It is advisable to use corticosteroids with caution in patients with ocular herpes simplex because of possible corneal perforation.
13. Psychic derangements may appear with corticosteroid therapy. Existing emotional instability or psychotic tendencies may be aggravated by corticosteroids.
14. Cautious use of corticosteroids in: non specific ulcerative colitis, if there is a probability of impending perforation, abscess, or other pyogenic infection, diverticulitis; fresh intestinal anastomoses; active or latent peptic ulcer, renal insufficiency, hypertension, osteoporosis and myasthenia gravis.
15. A risk/benefit decision must be made with each patient as complications of glucocorticoid treatment are dependent on dose size and duration of treatment.
16. Corticosteroids administration can disturb growth rates and inhibit endogenous corticosteroid production in infants and children, therefore the growth and development of these patients receiving prolonged therapy should be followed carefully.
17. The motility and number of spermatozoa may be altered in some patients receiving corticosteroids.

Interaction with Other Medicaments

1. Concurrent use of phenobarbital, phenytoin, rifampin or ephedrine may enhance the metabolism of corticosteroids, reducing their therapeutic effects.
2. Patients who are on both a corticosteroid and an estrogen should be observed for excessive corticosteroid effects.
3. The use of corticosteroids together with potassium-depleting diuretics may enhance hypokalemia. Concurrent use of corticosteroids with cardiac glycosides may enhance the possibility of arrhythmias or digitalis toxicity associated with hypokalemia. Corticosteroids may enhance the potassium depletion caused by amphotericin B. Serum electrolyte determinations, particularly potassium levels, should be monitored closely in all patients on any of these drug therapy combinations.
4. Adjustment of dosage may be required in patients taking corticosteroids and coumarin-type anticoagulants concurrently because this combination may increase or decrease the anticoagulant effects.
5. Concurrent use of non-steroidal anti-inflammatory drugs or alcohol with glucocorticosteroids may result in an increased occurrence or increased severity of gastrointestinal ulceration.
6. Since corticosteroids may decrease blood salicylate concentrations, acetylsalicylic acid should be used cautiously in conjunction with corticosteroids in hypoprothrombinemia.
7. Adjustment of dosage of an antidiabetic drug may be necessary when corticosteroids are given to diabetics.
8. Concomitant glucocorticosteroid therapy may inhibit the response to somatotropin.
9. Laboratory Test Interactions: Corticosteroids may affect the nitroblue tetrazolium test for bacterial infection and produce false-negative results.

Pregnancy and Lactation

The use of betamethasone tablets during pregnancy, in nursing mothers or women of child bearing age requires that the possible benefits of the drug be weighed against potential hazards to mother and foetus or infant because controlled human reproduction studies have not been done with corticosteroids.
Although no data are available for betamethasone, corticosteroids may pass into breast milk. Infants of mothers taking high doses of systemic corticosteroids for prolonged periods should be observed for signs of adrenal suppression.

Symptoms and Treatment for Overdosage, and Antidote(s):

Symptoms: Acute overdosage with betamethasone, like other glucocorticosteroids, is not expected to lead to a life-threatening situation. Except at the most extreme dosages, a few days of excessive glucocorticosteroid dosing is unlikely to produce harmful results in the absence of specific contraindications, such as in patients with diabetes mellitus, glaucoma, or active peptic ulcer, or in patients on medications such as digitalis, coumarin-type anti-coagulants or potassium-depleting diuretics
Treatment: There is no specific antidote for its overdosage, therefore management of the patients should consist of symptomatic and supportive therapy. Acute overdosage should be treated immediately by inducing emesis or by gastric lavage.

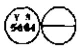
Shelf-Life:

3 years from the date of manufacture.

Storage Condition(s):

Keep in a tight container. Store at temperature below 30°C. Protect from light and moisture.

Product Description & Packing(s):

A light green color round tablet "  "

Plastic bottle of 1000's and 2000's.
Blister packing of 10's x 10.

Manufacturer: Y.S.P. INDUSTRIES (M) SDN. BHD.

Y. S. P.

Factory : Lot 3, 5 & 7, Jalan P/7, Section 13,
Kawasan Perindustrian Bandar Baru Bangi,
43000 Kajang, Selangor Darul Ehsan, Malaysia.
Tel: 03-89251215 (4 Lines) Fax: 03-89251298

Office : No. 18, Jalan Wan Kadir, Taman Tun Dr. Ismail,
60000 Kuala Lumpur, Malaysia.
Tel: 03-77276390 (12 Lines)
Ordering line:1 800 88 3027
Product info: 1 800 88 3679